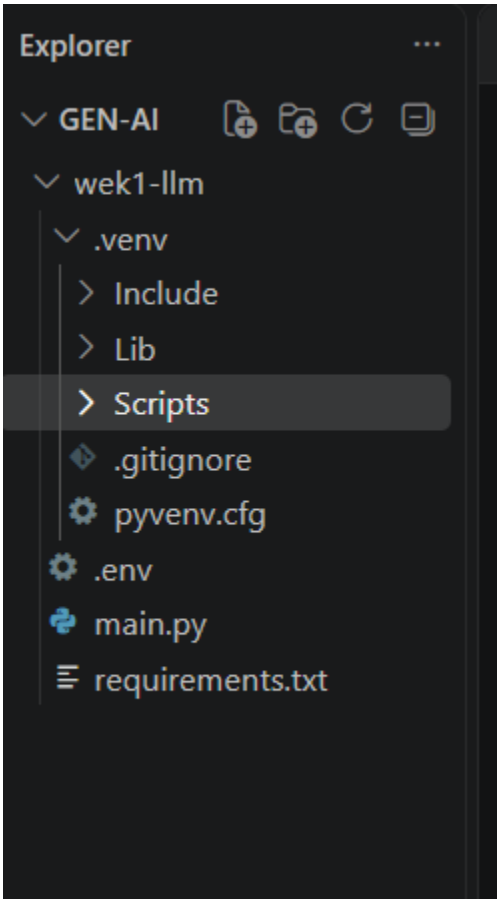
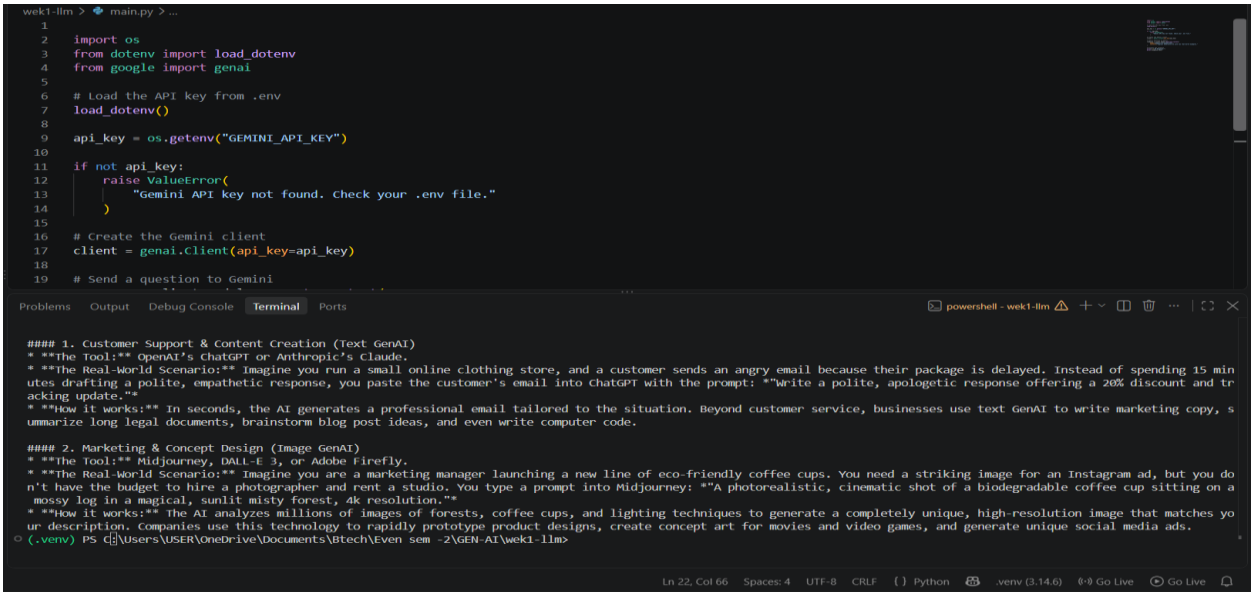
WEEK-1

2520090041

V.Pavani





main.py

```python
# Create the Gemini client
client = genai.Client(api_key=api_key)

# Send a question to Gemini
response = client.models.generate_content(
    model="gemini-3.5-flash-lite",
    contents="Differentiate Artificial Intelligence, Machine Learning, and Deep Learning."
)

# Display the response
print("\nLLM Response:")
print(response.text)
```

Problems Output Debug Console Terminal Ports

```
(.venv) PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\wek1-llm> python main.py
    *   Facial recognition software (on your phone or at airports).
    *   Generative AI tools like ChatGPT, Midjourney, and Google Translate.

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### Summary Comparison Table

| Feature | Artificial Intelligence (AI) | Machine Learning (ML) | Deep Learning (DL) |
| :--- | :--- | :--- | :--- |
| **Scope** | Broadest (The concept) | Subset of AI | Subset of ML |
| **Primary Goal** | Simulate human intelligence | Learn from data to make predictions | Handle complex data using neural networks |
| **Human Intervention** | High (often requires hardcoded rules) | Medium (requires humans to define features) | Low (automatically extracts features) |
| **Data Requirements** | Can work with little to no data (rules-based) | Requires moderate to large amounts of data | Requires **massive** amounts of data |
| **Hardware Needed** | Standard computers | Standard computers | High-powered GPUs (Graphics Processing Units) |
(.venv) PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\wek1-llm>
```

main.py requirements.txt

```python
# Create the Gemini client
client = genai.Client(api_key=api_key)

# Send a question to Gemini
response = client.models.generate_content(
    model="gemini-3.5-flash-lite",
    contents="What is a Large Language Model? Explain for a first-year engineering student."
)

# Display the response
print("\nLLM Response:")
print(response.text)
```

Problems Output Debug Console Terminal Ports

```
(.venv) PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\wek1-llm> python main.py

To turn it into an assistant, models go through two more phases:
1.  **Supervised Fine-Tuning (SFT):** Training the model on high-quality examples of *Instructions* and *Responses*.
2.  **Reinforcement Learning from Human Feedback (RLHF):** Having humans (and other AI models) rank the LLM's outputs, using reinforcement learning to steer the model toward being helpful, honest, and harmless.

### Summary for an Engineer
*   **It's not a database:** It doesn't "look up" facts. It compresses internet-scale data into statistical relationships between mathematical vectors.
*   **It's not sentient:** It has no internal model of the world; it has a remarkably sophisticated model of *language syntax and semantics*.
*   **It is probabilistic, not symbolic:** It doesn't use formal logic trees to solve problems; it calculates high-probability trajectories through a massive vector space.

As an engineering student, you'll find that LLMs are powerful tools—not just for you to use in your daily workflow, but fascinating complex systems sitting at the intersection of linear algebra, optimization, probability, and computer architecture.
(.venv) PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\wek1-llm>
```

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